10(a)(i)	total power	**B1**
	power radiated (by the star)	**B1**
10(a)(ii)	standard candle has known luminosity	**B1**
	radiant flux intensity measured by observer	**B1**
	(distance calculated using) $F = L / 4\pi d^2$	**B1**
10(b)(i)	luminosity $= 4\pi\sigma r^2 T^4$ $= 4\pi \times 5.67 \times 10^{-8} \times (6.96 \times 10^8)^2 \times 5780^4$	**C1**
	$= 3.85 \times 10^{26}$ W	**A1**
10(b)(ii)	$\lambda_{MAX}T$ = constant	**C1**
	temperature $= (5780 \times 501) / 624$ $= 4640$ K	**A1**